Topological Data Analysis — Lecture 2

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AGQ Computing Course

6 March 2026

Recap of Persistent Homology (PH)

Algebra Geometry & Quantum Fields

Topological Data Analysis (TDA) refers to statistical methods that rely on topological ideas to find structure in data

Persistent homology pipeline

1. Metric geometry: Given data $X = \{X_1, \ldots, X_n\} \subset M$ in some metric space, construct a family of cell complexes $\{X_t \mid t \in \mathbb{R}_{\geq 0}\}$
2. Algebraic topology: Compute the *k*-th homology vector spaces $H_k(X_t)$ across scales $t \in \mathbb{R}_{\geq 0}$ and dimensions $k \in \mathbb{Z}_{\geq 0}$
3. Representation theory: Decompose each family of vector spaces $\{H_k(X_t) \mid t \geq 0\}$ into irreducible summands yielding *barcodes*

The barcodes are finite multi-sets of real intervals $[p, q] \subset \mathbb{R}_{\geq 0}$

Recap: Filtered simplicial complexes

Algebra
Geometry &
Quantum Fields

Main example. Finite subset of points $M = \{x_0, x_1, \ldots, x_n\} \subset \mathbb{R}^d$

Definition. Let (M, d) be a finite metric space. The Vietoris-Rips filtration is an "increasing family" of simplicial complexes $\mathrm{VR}_\epsilon(M)$ indexed by the real numbers $\epsilon \geq 0$ defined as follows:

A subset $\{x_0, x_1, \ldots, x_k\} \subset M$ forms a k-dimensional simplex in $\mathrm{VR}_\epsilon(M)$ if and only if we have $d(x_i, x_j) \leq \epsilon \text{ for all } i, j$

Recap: Persistent homology groups

With a filtered simplicial complex in place (either VR or Cech), we obtain a persistence module from the k-th homology, i.e.,

$$H_k(F_\bullet K)\colon H_k(F_0 K) \to H_k(F_1 K) \to \ldots \to H_k(F_{n-1} K) \to H_k(F_n K)$$

we obtain persistent homology groups $\mathrm{PH}_k\, g_{i\to j}(F_\bullet K)$ for $i \leq j$

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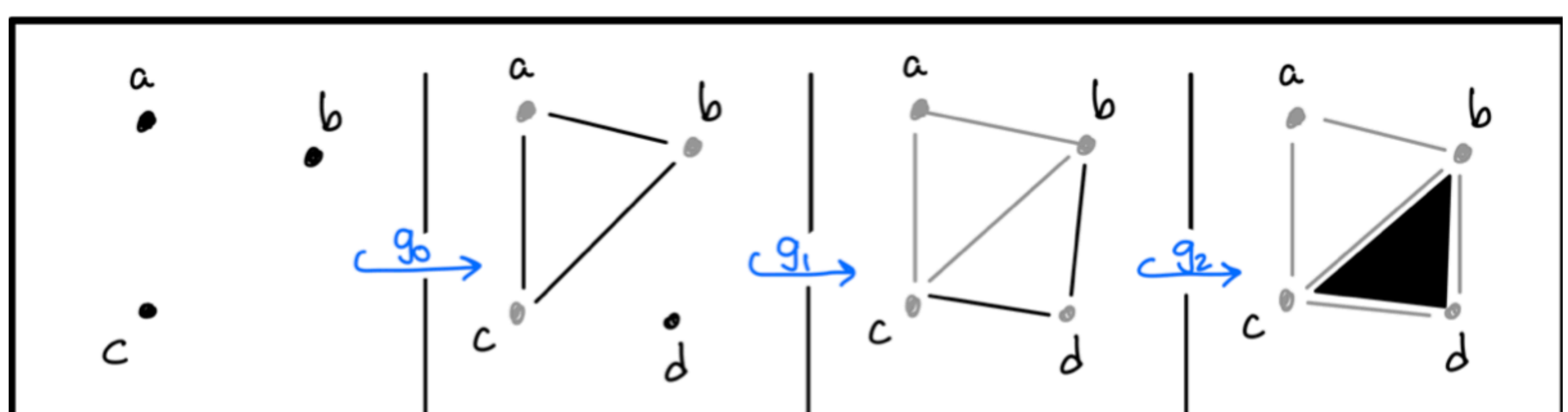
Interpretation. The elements are homology classes in $H_k(F_iK)$ which continue to generate non-trivial homology classes in the larger simplicial complex F_jK. These are (equivalence classes of) k-cycles in F_iK which do not become k-boundaries in F_jK

Compare

$\{a, b, c\} \in \mathrm{PH}_1\, g_{1\to 3}(F_\bullet K)$
$\{b, c, d\} \in \mathrm{PH}_1\, g_{2\to 3}(F_\bullet K)$



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In fact: $$\mathrm{PH}_k\, g_{i\to j}(F_\bullet K) = \frac{H_k g_{i\to j}\left(\ker \partial_k^i\right)}{H_k g_{i\to j}\left(\ker \partial_k^i\right) \cap \mathrm{im}\left(\partial_{k+1}^j\right)}$$

Note that $\mathrm{PH}_k\, g_{i\to i}(F_\bullet K) = H_k\left(F_i K\right)$

Compare

$\{a, b, c\} \in \mathrm{PH}_1\, g_{1\to 3}(F_\bullet K)$
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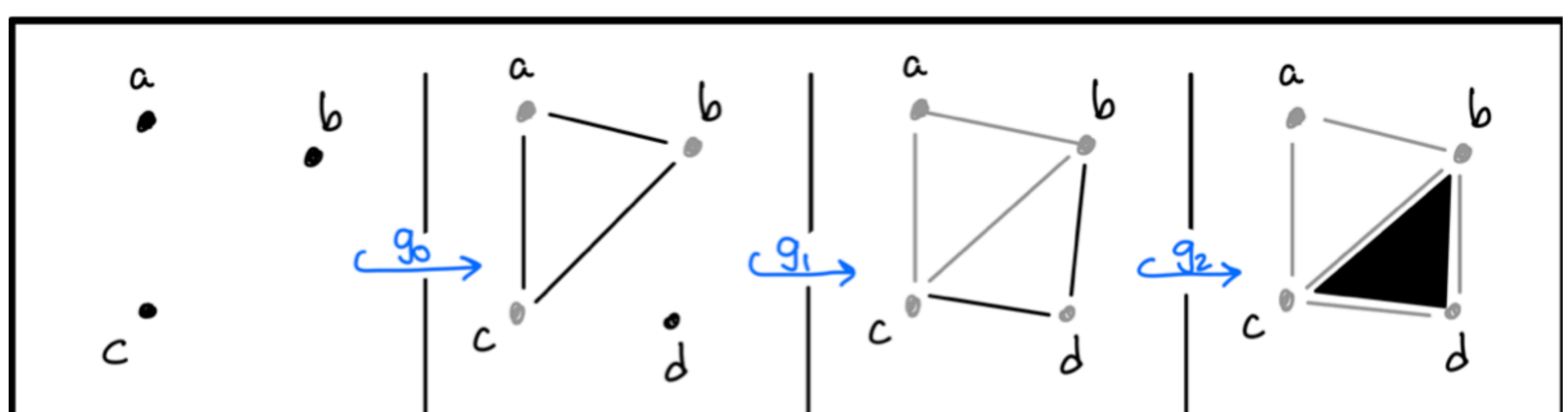


Recap: Barcode theorem

Algebra Geometry & Quantum Fields

finite type,

Theorem. For any $\mathbb{N}$-indexed persistence module $(V_\bullet, a_\bullet)$ over $\mathbb{F}$ there exists a set $\mathrm{Bar}(V_\bullet, a_\bullet)$ of integer pairs $0 \leq i \leq j \leq \infty$ and a function $\mu \colon \mathrm{Bar}(V_\bullet, a_\bullet) \to \mathbb{N}_{>0}$ with the following property: There is a direct sum decomposition

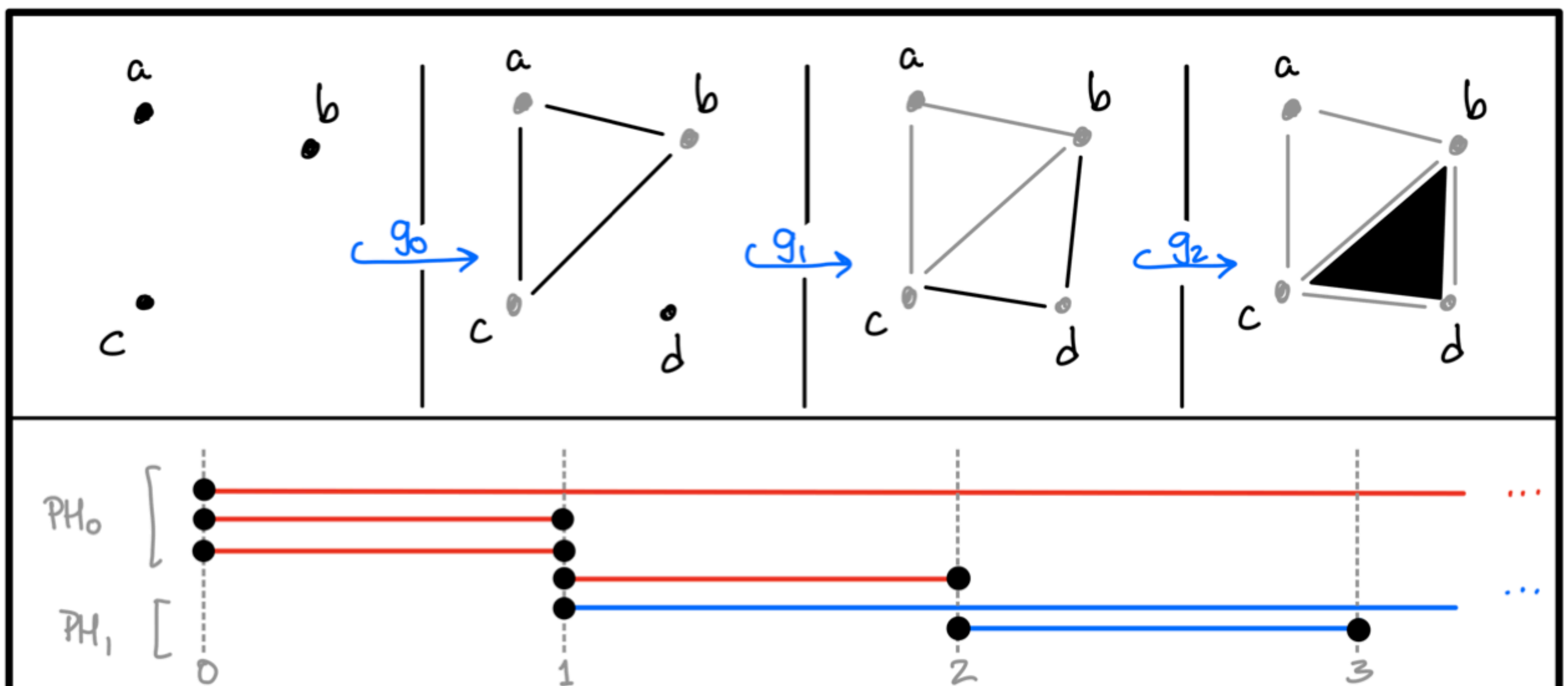
$$(V_\bullet, a_\bullet) \simeq \bigoplus_{[i,j]} \left(I_\bullet^{i,j}, c_\bullet^{i,j}\right)^{\mu(i,j)}$$

which is unique up to isomorphism of persistence modules.

Recap: Barcode theorem

Definition. For any $\mathbb{N}$-indexed persistence module $(V_\bullet, a_\bullet)$ over $\mathbb{F}$ of finite type, the collection $\mathrm{Bar}(V_\bullet, a_\bullet)$ of intervals $[i,j]$ and their multiplicities $\mu(i,j) \geq 1$ is called the barcode of $(V_\bullet, a_\bullet)$

Example.



Ch. 6, *Computational Algebraic Topology* by Vidit Nanda

Structure of the lectures

Lecture 1 (last week): Computing persistent homology

- Simplicial complexes and simplicial homology
- Vietoris-Rips complex and computation of PH
- Structure result: The barcode theorem

Tutorial: Pen-and-paper questions + notebook to compute PH

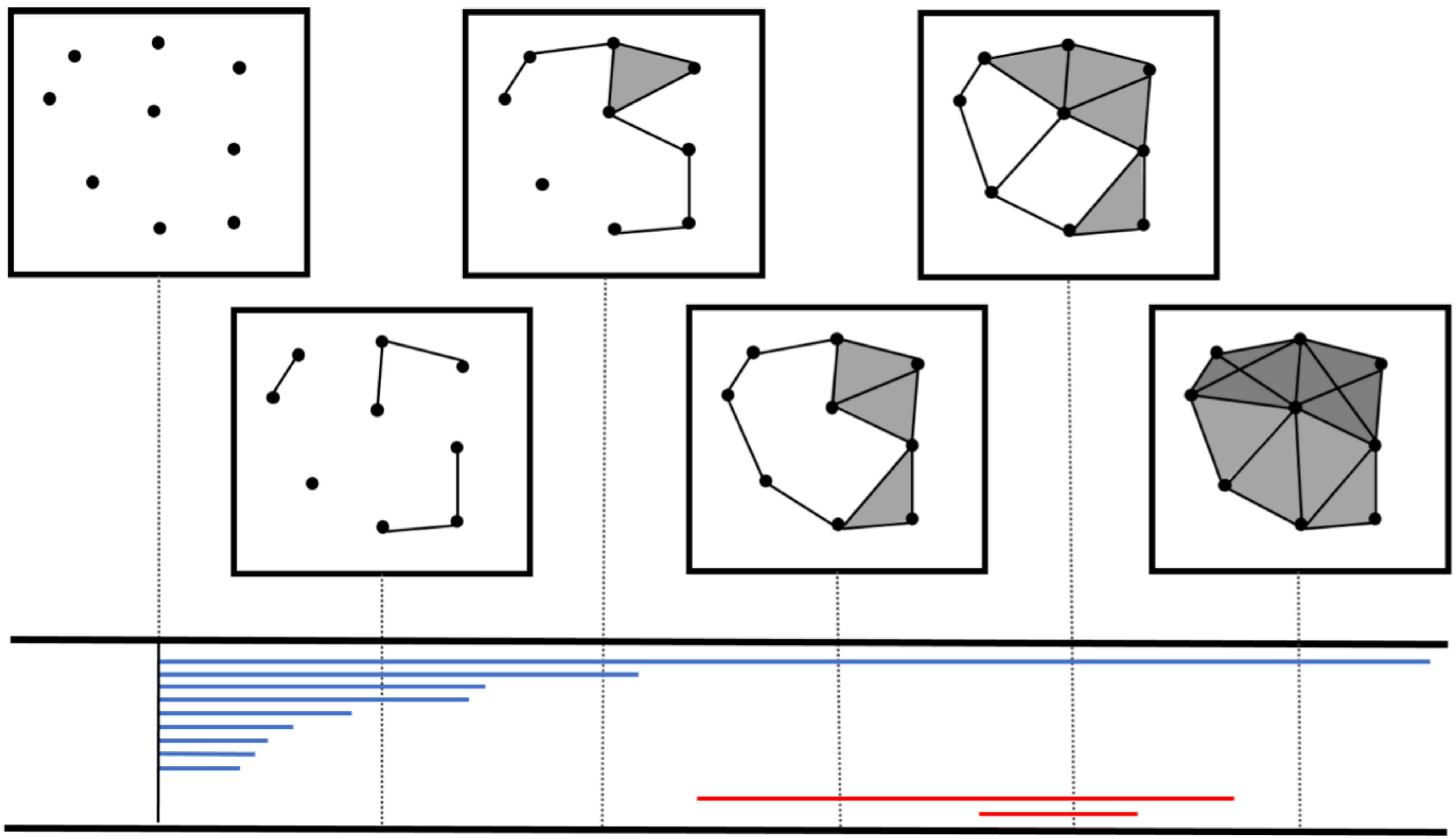
Lecture 2 (today): Leveraging persistent homology

- Stability of barcodes and persistence diagrams
- Vectorisation of persistence diagrams
- Applications of PH/TDA in machine learning

Tutorial: Data sets to sharpen your skills!

Barcode theorem (revisited)

Remark. Our persistence modules are $\mathbb{R}_{\geq 0}$-indexed



Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Barcode theorem (revisited)

Algebra Geometry & Quantum Fields

Remark. Our persistence modules are $\mathbb{R}_{\geq 0}$-indexed

Definition. An $\mathbb{R}_{\geq 0}$-indexed **persistence module** over $\mathbb{F}$ is a pair $(V_\bullet, a_\bullet)$ of vector spaces $\{V_t\}$ and linear maps $\{a_{s\leq t}\colon V_s \to V_t\}$ s.t.

1. Identity: $a_{t\leq t}$ is the identity on V_t
2. Composition: $a_{s\leq t} \circ a_{r\leq s} = a_{r\leq t}$ for all $0 \leq r \leq s \leq t$

Note. A persistence module is a functor $(\mathbb{R}_{\geq 0}, \leq) \to \mathbf{Vect}_{\mathbb{F}}$

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Note. A persistence module is a functor $(\mathbb{R}_{\geq 0}, \leq) \to \mathbf{Vect}_{\mathbb{F}}$

Definition. A persistence module $(V_\bullet, a_\bullet)$ is called **tame** if

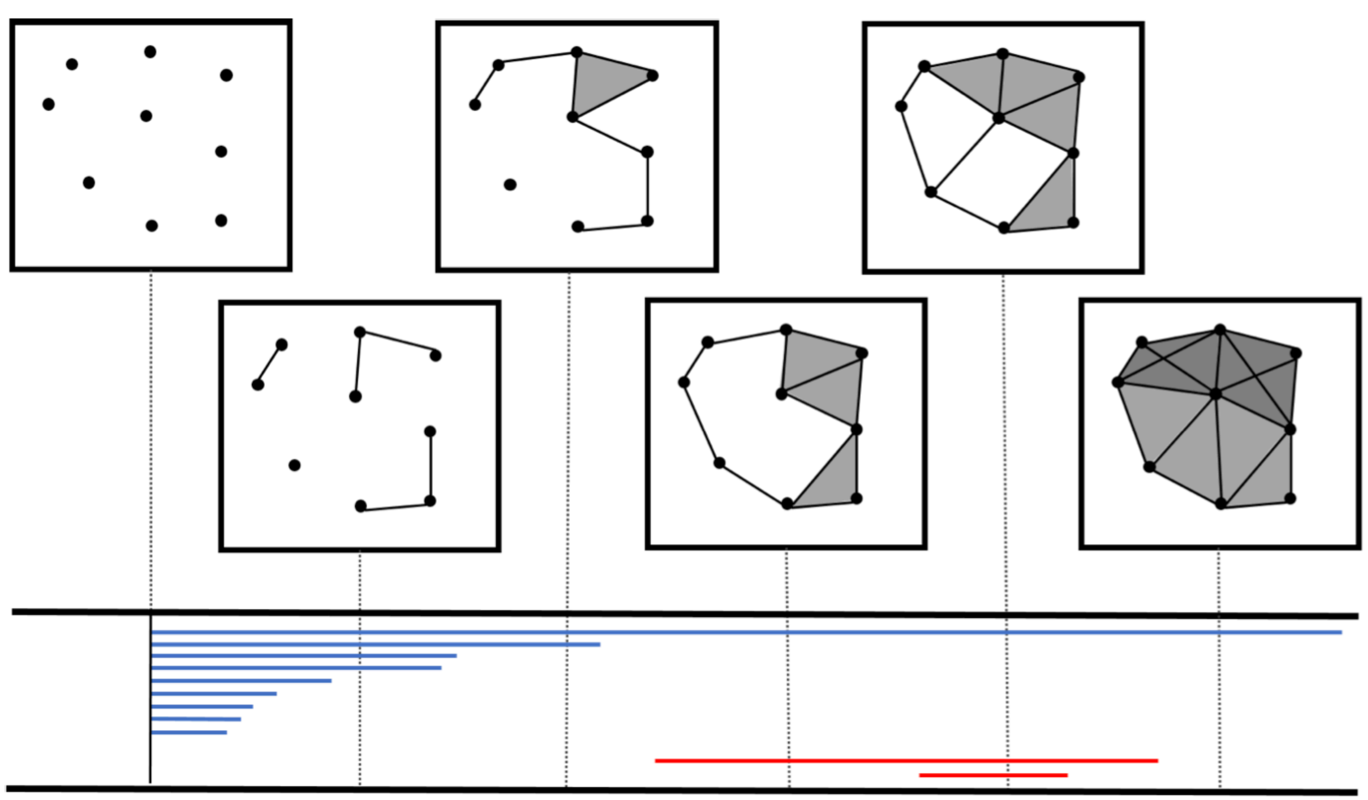
1. The V_t are finite-dimensional
2. There are only finitely many $t \geq 0$ s.t. $a_{t-\epsilon, t+\epsilon} \colon V_{t-\epsilon} \to V_{t+\epsilon}$ fails to be an isomorphism for arbitrarily small $\epsilon > 0$

Barcode theorem (revisited)

Theorem. For every tame persistence module $(V_\bullet, a_\bullet)$ over $\mathbb{F}$ there exists a set $\mathrm{Bar}(V_\bullet, a_\bullet)$ of intervals $[s,t] \subset \mathbb{R}_{\geq 0}$ and a function $\mu\colon \mathrm{Bar}(V_\bullet, a_\bullet) \to \mathbb{N}_{>0}$ with a $\oplus$-decomposition

$$(V_\bullet, a_\bullet) \simeq \bigoplus_{[s,t]} \left(I_\bullet^{s,t}, c_\bullet^{s,t}\right)^{\mu(s,t)}$$

which is unique up to isomorphism of persistence modules.



Ch. 6, *Computational Algebraic Topology* by Vidit Nanda
Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Stability theorem

By the Barcode theorem, we have an identification of

- Tame persistence modules $(V_\bullet, a_\bullet)$ (up to isomorphism)
- Multi-sets of intervals $[s,t] \subset \mathbb{R}_{\geq 0} \cup \{\infty\}$

so an algebraic object and combinatorial data

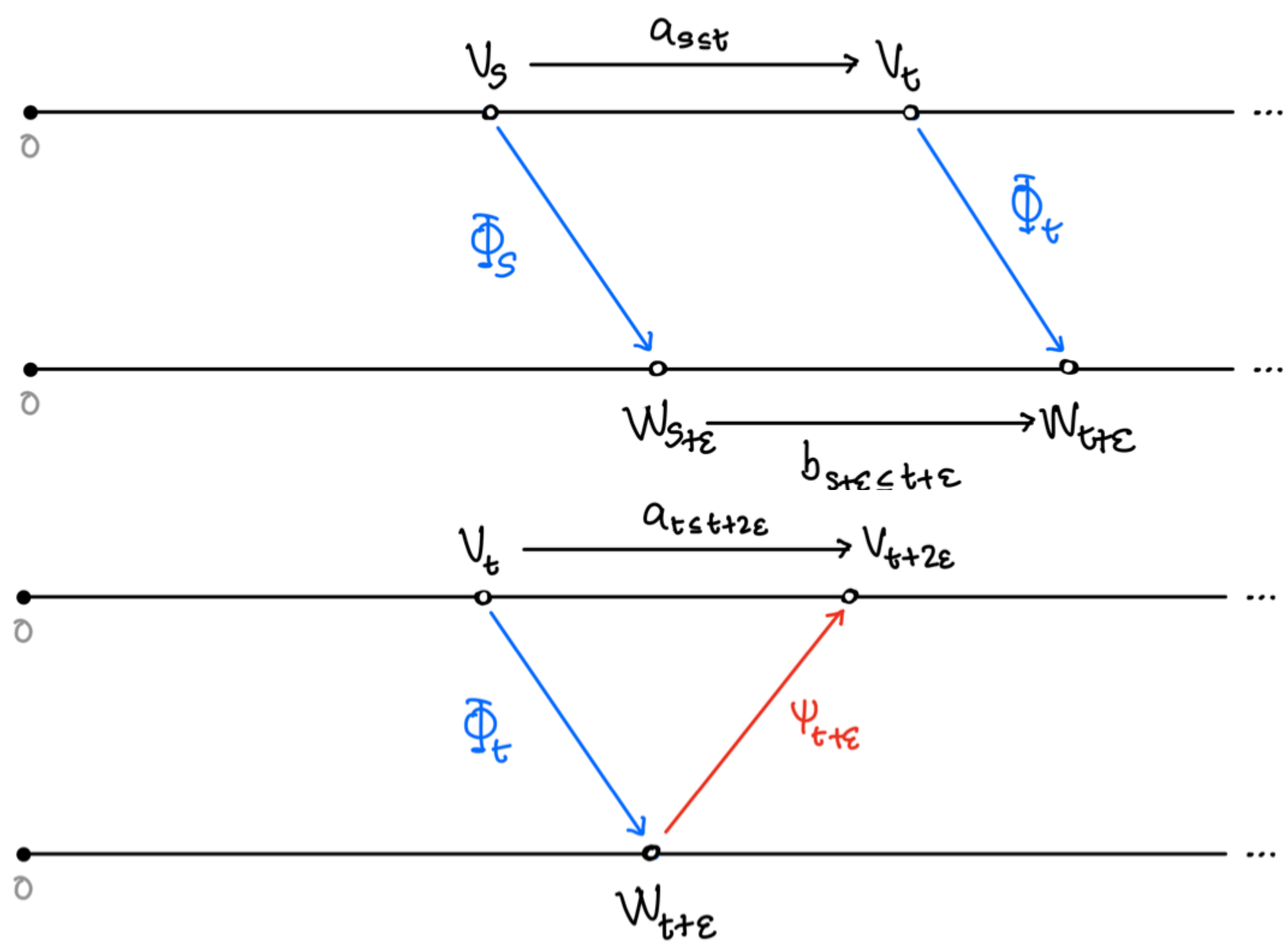
There are natural metrics on both sides:

- *Interleaving distance*, on the set of persistence modules
- *Bottleneck distance*, on the set of barcodes

Stability theorem

The assignment $(V_\bullet, a_\bullet) \mapsto \mathrm{Bar}(V_\bullet, a_\bullet)$ is an isometry

Stability theorem: Interleaving



Definition. The *interleaving distance* between persistence modules is the infimum over all $\epsilon \geq 0$ for which there is an ϵ-interleaving between them. If no such interleaving exists, it is infinite

Note. 0-interleavings are isomorphisms of persistence modules

Stability theorem: Bottleneck

Definition. An ϵ-matching between multi-sets of intervals is a bijection $\rho\colon B_0 \to B_0'$ of multi-subsets $B_0 \subset B, B_0' \subset B'$ s.t.

1. Every $[s,t] \in (B \setminus B_0) \cup (B' \setminus B_0')$ has length $t - s \leq 2\epsilon$
2. If $\rho[s,t] = [s',t']$ then we have $|s - s'|, |t - t'| \leq \epsilon$

Idea: All "long" intervals are paired up with intervals "close by"

Definition. The *bottleneck distance* between multi-sets of intervals is the infimum over all $\epsilon \geq 0$ for which there is an ϵ-matching between them.

Ch. 6, *Computational Algebraic Topology* by Vidit Nanda

Application of the Stability theorem

Algebra Geometry & Quantum Fields

Proposition. Let P and Q be two finite point-sets in $\mathbb{R}^n$ which are close in the following sense. There is some $\epsilon > 0$ for which

1. There is a point of Q within distance ϵ of any point in P,
2. There is a point of P within distance ϵ of any point in Q.

Then for $k \geq 0$, the k-th persistence homology modules of the Vietoris-Rips filtrations $\mathrm{VR}_\bullet(P)$ and $\mathrm{VR}_\bullet(Q)$ are 2ϵ-interleaved

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Corollary. For any finite point-sets $P, Q \subset \mathbb{R}^n$ as above, the k-th order PH barcodes from the VR complex have the same number of "sufficiently long" bars, i.e., those with length $\geq 4\epsilon$.

Question: Points on a circle with(out) outlier point in the middle?

Ch. 6, *Computational Algebraic Topology* by Vidit Nanda

Algebra Geometry & Quantum Fields

Persistent homology in Machine Learning

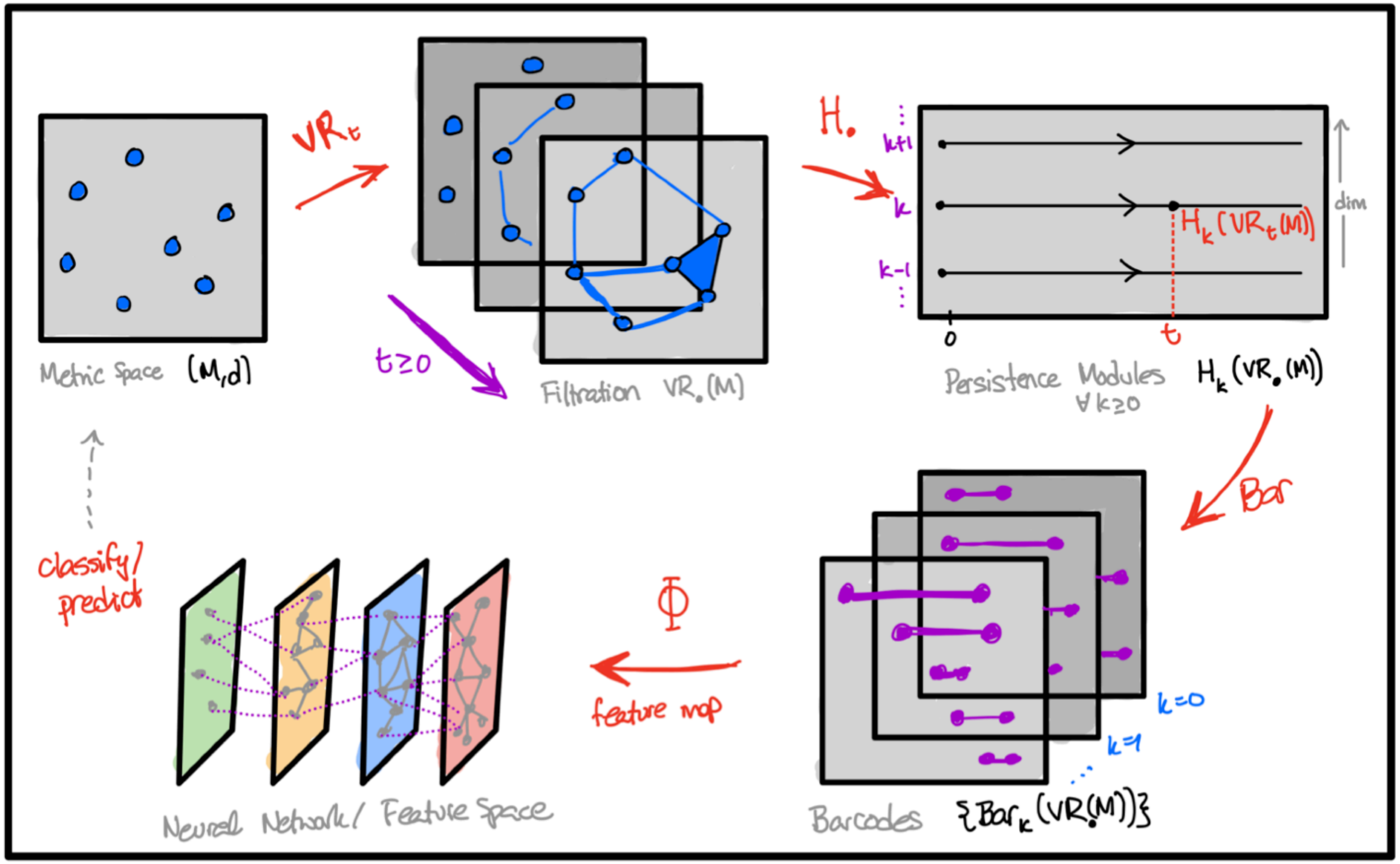

Figure from *Computational Algebraic Topology* by Vidit Nanda

Leveraging PH in ML

Algebra Geometry & Quantum Fields

Almost completed the *topological data analysis pipeline*: what remains is preparing the multi-set of barcodes for a training task

For unstructured data without an inner product, use the kernel trick, i.e., embed data into a Hilbert space of functions

$$\phi\colon \mathcal{B} \to \mathcal{H}_{\mathcal{B}} \subset \{f : \mathcal{B} \to \mathbb{R}\}$$

via the feature map. By Moore-Aronszajn (1950), equivalent to the choice of a symmetric positive definition kernel on $\mathcal{B}$

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Despite nice isometry between persistent modules and barcodes, these metric spaces do not have favourable properties

Theorem (Bubenik-Wagner, 2020). The metric space of barcodes with the bottleneck distance does not admit an isometric (or even coarse) embedding into a Hilbert space.

Leveraging PH in ML

Algebra Geometry & Quantum Fields

Coarse embeddings are a more general notion of embedding:

1. Not necessarily continuous,
2. Distances need to be distorted in a uniform way, which is allowed to be non-linear.

Conclusion. Any vectorisation of the metric space of persistence modules (or barcodes), i.e., any possible feature map into a Hilbert space, necessarily distorts the original metric

… *thus the method of vectorisation is a choice, analogous to setting the value of hyper-parameters in ML*

Bubenik-Wagner (2020), Embeddings of persistence diagrams into Hilber spaces

Leveraging PH in ML

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Note. It is possible to work with distance between PMs or barcodes only, but this gets computationally costly!

Bubenik-Wagner (2020), Embeddings of persistence diagrams into Hilber spaces

Overview of vectorisations

Based on *A Survey of Vectorisation Methods in TDA (2023)*

They distinguish 5 (non-disjoint) classes of vectorisations

1. *Statistical*: A vector of basic statistical quantities
2. *Algebraic*: Generated from polynomials
3. *Curve*: Maps into a vector space of the form $\mathbb{R} \to H$
4. *Functional*: Similar, but $\mathbb{R} \neq X \to H$
5. *Ensemble*: Generated from collections of training barcodes

Compare performance in three classification tasks

Provide a web (+ downloadable) app to play with!

Curve vectorisation

Definition. Given the barcode $B = \mathrm{Bar}(V_\bullet, a_\bullet)$ of a tame PM with multiplicity $\mu\colon B \to \mathbb{Z}_{>0}$ and an interval [p,q], we call p the *birth*, q the *death*, and the length q-p the *lifespan* of the bar, respectively.

Definition. The Betti curve of $\mu\colon B \to \mathbb{Z}_{>0}$ is $\beta_\mu\colon \mathbb{R}_{\geq 0} \to \mathbb{R}$

$$\beta_\mu(t) = \sum_{[p,q]\in B} \mathbb{1}_{p\leq t\leq q} \cdot \mu_{p,q}$$

counting the number of intervals (+ multiplicity) which contain t.

Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

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Algebra Geometry & Quantum Fields

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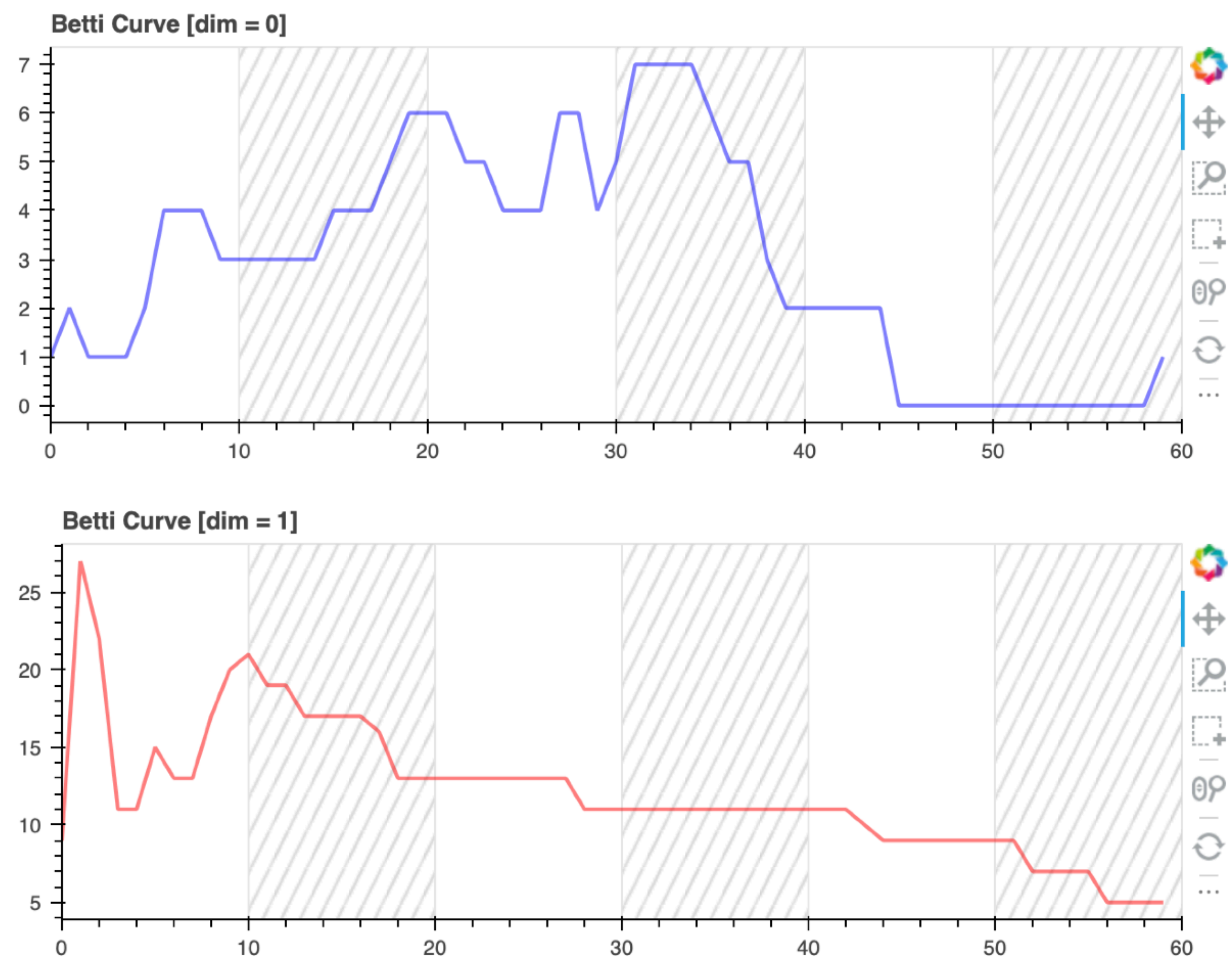
Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Curve vectorisation: Betti

In practice, we choose, e.g., 100 equally spaced values of the parameter t, and record the function as an object in $\mathbb{R}^{100}$.



Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Curve vectorisation: Lifespan

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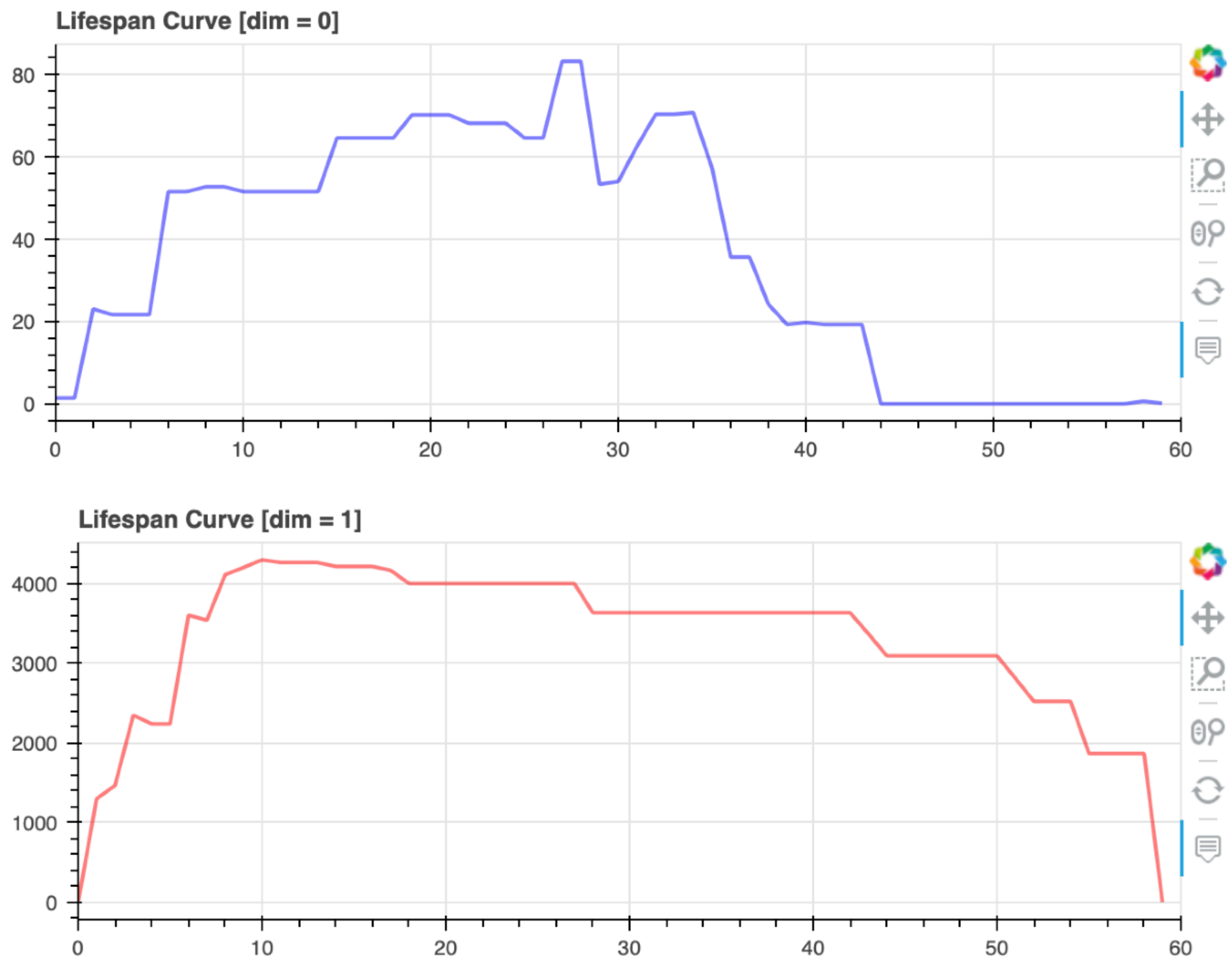

Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Curve vectorisation: Downside

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Definition. The lifespan curve of $\mu\colon B \to \mathbb{Z}_{>0}$ is $L_\mu : \mathbb{R}_{\geq 0} \to \mathbb{R}$

$$L_\mu(t) = \sum_{[p,q]\in B} \mathbb{1}_{p\leq t\leq q} \cdot \mu_{p,q} \cdot (q-p)$$

Downside: These notions are not "stable". By adding lots of very small intervals, we can make the curves very different, yet make sure they have a small bottleneck distance

Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Curve vectorisation: Landscapes

Persistent landscapes offer a stabilisation, in which we replace the condition $t \in [p,q] \in B$ with the largest $s \geq 0$ such that

$$[t-s, t+s] \in [p,q] \in B$$

Definition. The persistence landscape of $\mu\colon B \to \mathbb{Z}_{>0}$ is a sequence of curves $\Lambda_i^\mu\colon \mathbb{R}_{\geq 0} \to [-\infty, \infty]$ for $i \in \mathbb{Z}_{>0}$ with image

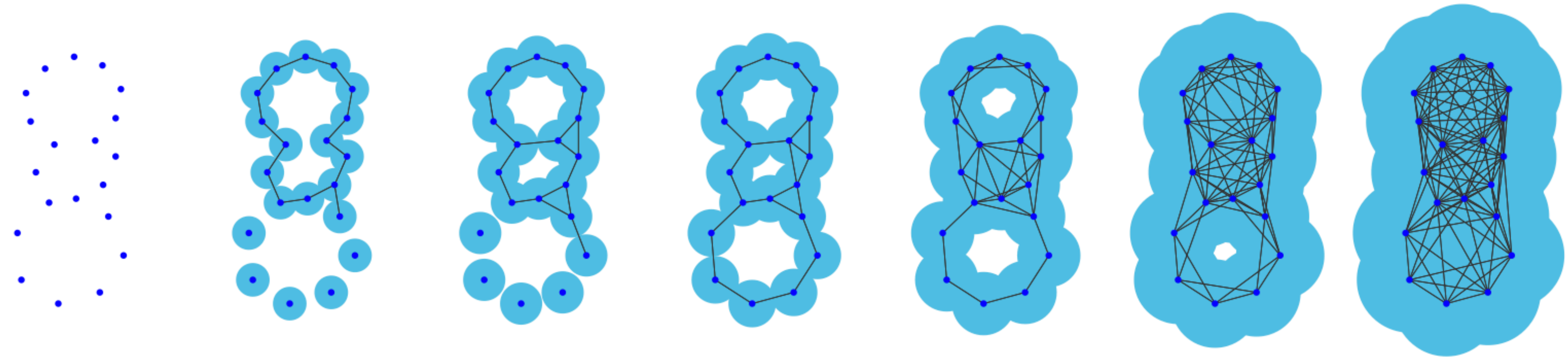
$$\Lambda_i^\mu(t) = \sup\left\{ s \geq 0 \mid \sum_{[p,q]\in B} \mathbb{1}_{[t-s,t+s]\subset[p,q]} \cdot \mu_{p,q} \geq i \right\}$$

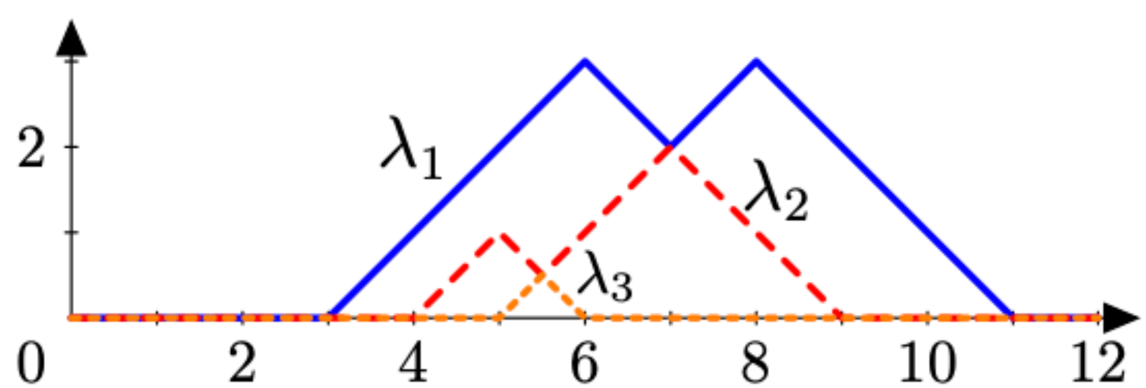
Remarks.

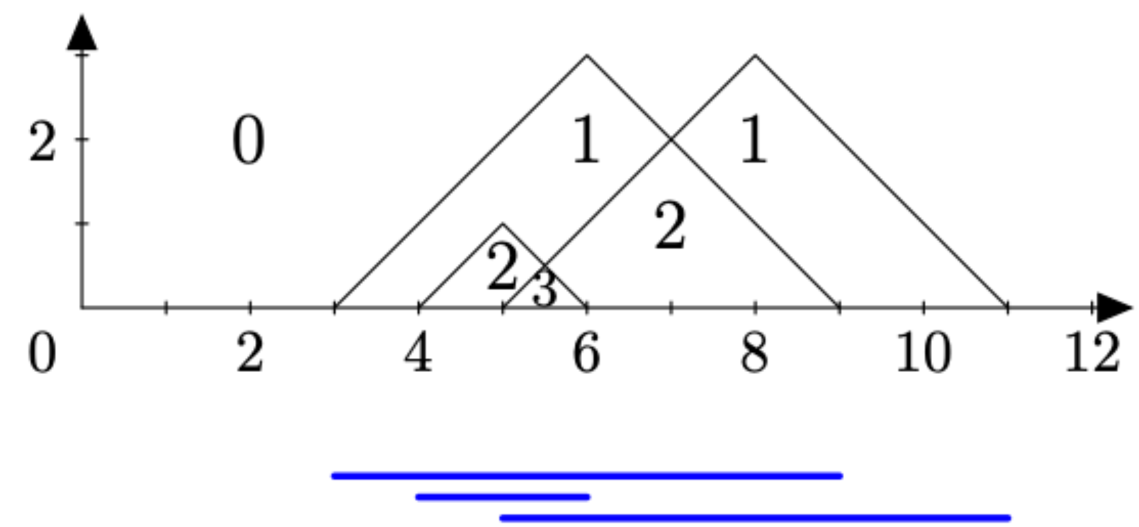
1. The supremum over the empty set is zero
2. Since our barcode B is finite, the functions are zero for $i \gg 0$

Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Curve vectorisation: Landscapes

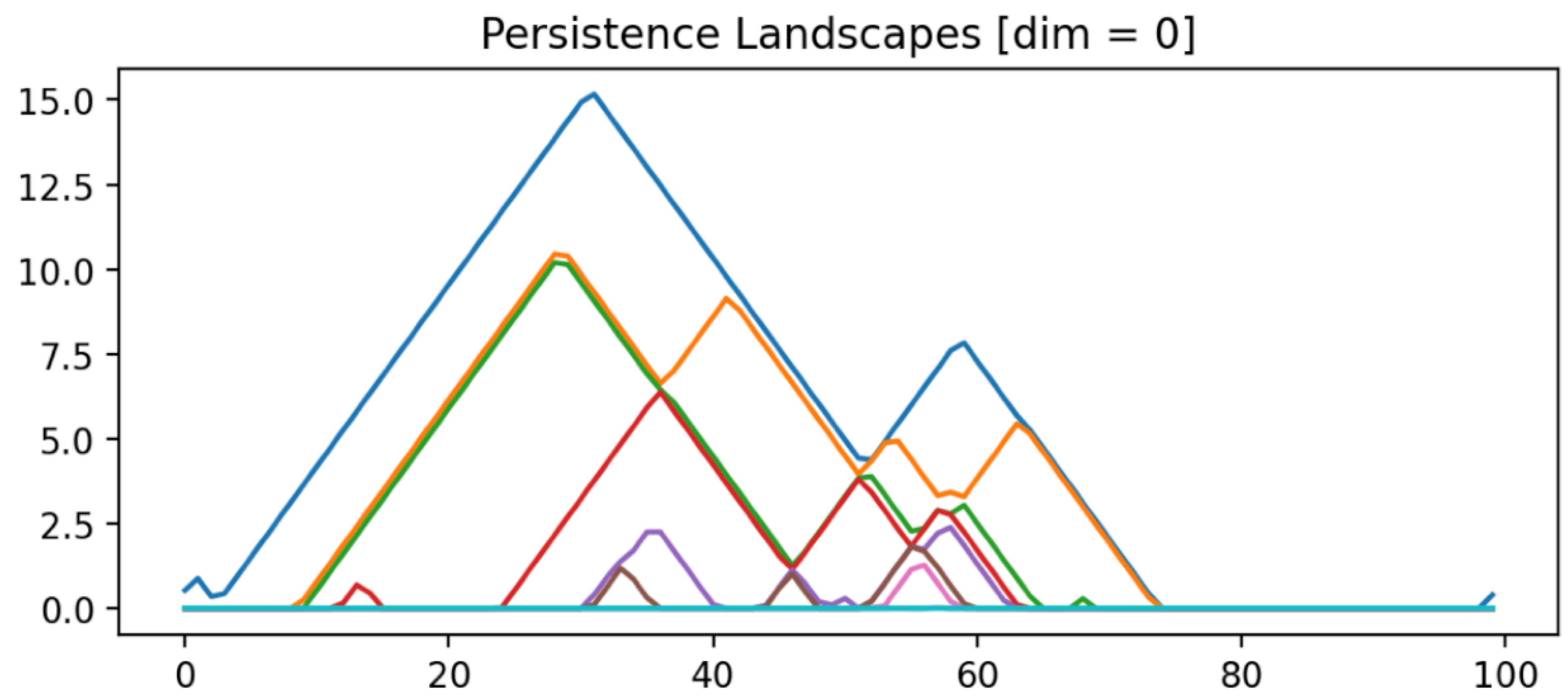


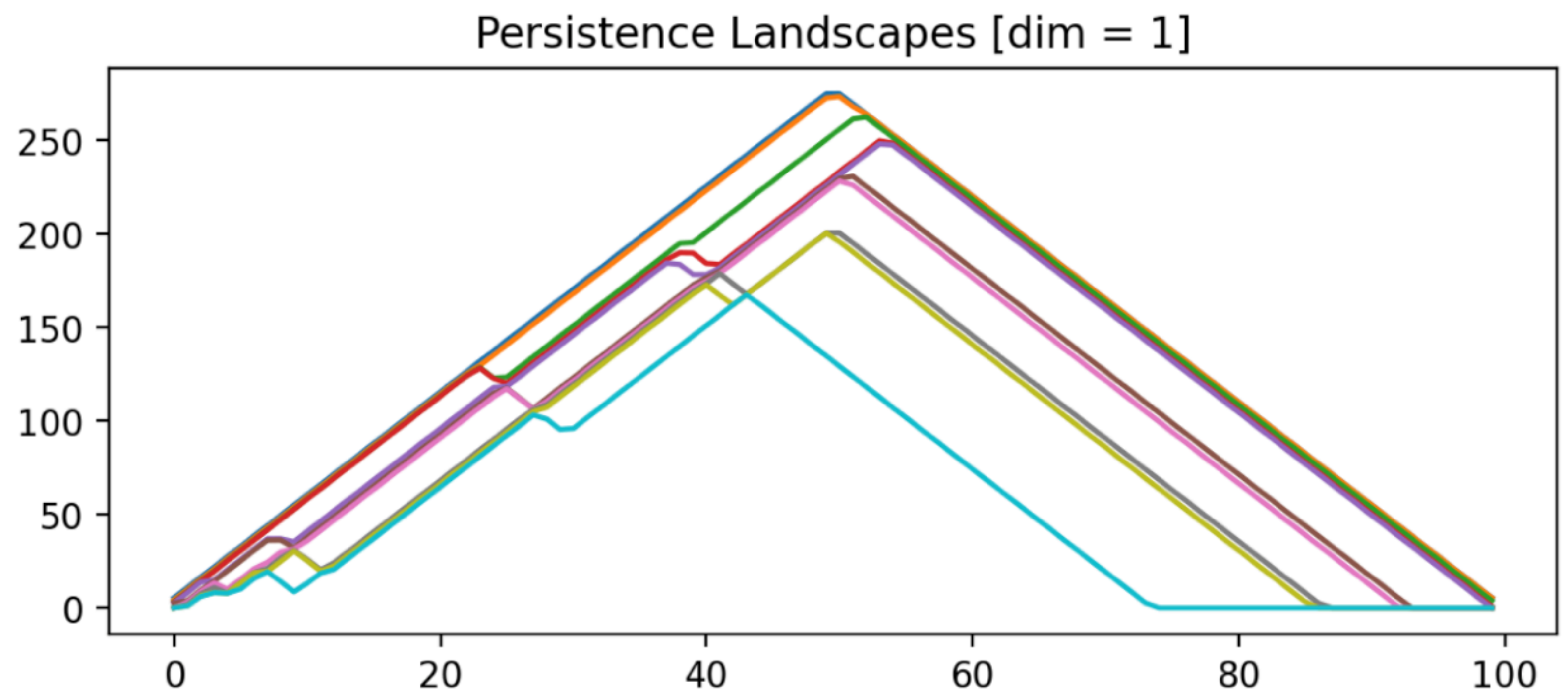




Bubenik (2015), Statistical TDA using Persistence Landscapes

Curve vectorisation: Landscapes





Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Statistics on Persistence Landscapes

Algebra Geometry & Quantum Fields

Via the vector space structure, we can take means of landscapes

Suppose we sample n point clouds, each yielding $\mu_j : B_j \to \mathbb{Z}_{>0}$

We obtain a *mean persistence landscape* via

$$\overline{\Lambda}_i(t) = \frac{1}{n} \sum_{j=1}^{n} \Lambda_i^{\mu_j}(t)$$

Bubenik (2015), Statistical TDA using Persistence Landscapes

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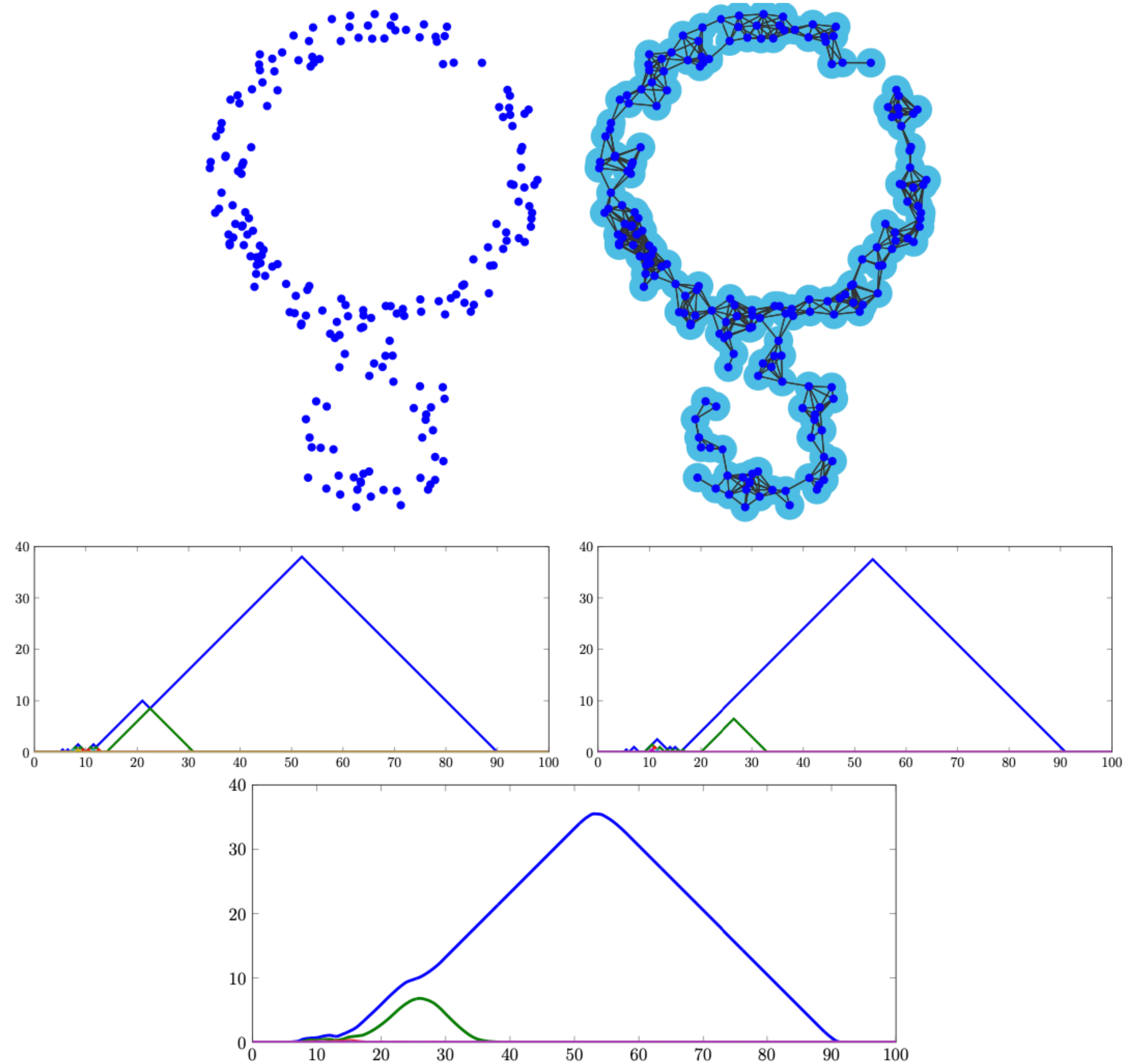
While this need not correspond to a meaningful landscape, using some probability theory on Banach spaces and obtain:

1. Strong LLN: $\overline{\Lambda}_i \to \mathbb{E}[\Lambda_i]$ almost surely
2. CLT: $\sqrt{n}\left(\overline{\Lambda}_i - \mathbb{E}[\Lambda_i]\right)$ converges weakly to a Gaussian

By applying a functionals to PL's we obtain a regular (real-valued) random variable + CLT + hypothesis testing!

Bubenik (2015), Statistical TDA using Persistence Landscapes

Statistics on Persistence Landscapes



Bubenik (2015), Statistical TDA using Persistence Landscapes

Use in ML

Algebra Geometry & Quantum Fields

After selecting an appropriate vectorisation of our barcodes, we can next incorporate our newly defined *topological features* in a loss function of interest for a given machine learning task

Vectorization Method	Score	Param.	Estimator
Persistence Statistics	**0.947**	t_5	RF, n=100
Entropy Summary	0.723	t_6, 200	RF, n=300
Algebraic Functions	0.909	t_5	RF, n=500
Tropical Coordinates	0.844	t_6, 50	SVM, lin. kernel, C_5
Complex Polynomial	0.889	t_6, 20, S	SVM, lin. kernel, C_6
Betti Curve	0.728	t_5, 200	RF, n=100
Lifespan Curve	0.878	t_7, 200	SVM, lin. kernel, C_7
Persistence Landscape	0.889	t_6, 50, 10	SVM, rbf kernel, C_1, γ_1
Persistence Silhouette	0.867	t_6, 200, 2	SVM, rbf kernel, C_2, γ_2
Persistence Image	0.916	t_5, 1, 10	RF, n=100
Template Function	0.944	t_5, 14, 0.7	SVM, rbf kernel, C_3, γ_3
Adaptive Template Sys.	0.889	t_5, GMM, 15	SVM, lin. kernel, C_8
ATOL	0.933	t_8, 16	SVM, rbf kernel, C_4, γ_4

TABLE 3

Best performance of each method on **SHREC14**. The parameters are $C_1 = 835.6257$, $\gamma_1 = 0.0002$, $C_2 = 212.6281$, $\gamma_2 = 0.0031$, $C_3 = 879.1425$, $\gamma_3 = 0.0010$, $C_4 = 936.5391$, $\gamma_4 = 0.0187$, $C_5 = 141.3869$, $C_6 = 625.0300$, $C_7 = 998.1848$, $C_8 = 274.500$.

Task: Classification (15 classes), 70/30 training/testing, Score = accuracy

Ali et al (2023), A Survey of Vectorisation Methods in TDA, IEEE

Use in ML

Algebra Geometry & Quantum Fields

After selecting an appropriate vectorisation of our barcodes, we can next incorporate our newly defined *topological features* in a loss function of interest for a given machine learning task

Approach. Add a topological term to the loss function

$$\mathcal{L} = \mathcal{L}_0 + \lambda \mathcal{L}_{\text{topo}}$$

with a parameter to tune the relative contribution

Training. Differentiability a.e. of the persistence map, and thus for persistence based functions that are locally Lipschitz (e.g., PL) — required for back-propagation and stochastic gradient descent

Carrière et al (ICML 2021), Optimising PH based functions

Topological Data Analysis — Lecture 2

Sjoerd Beentjes & Siddharth Setlur

AGQ Computing Course

6 March 2026

Recap: Filtered simplicial complexes

Main example. Finite subset of points $M = \{x_0, x_1, \ldots, x_n\} \subset \mathbb{R}^d$

Definition. Let (M, d) be a finite metric space. The Vietoris-Rips filtration is an “increasing family" of simplicial complexes $\mathrm{VR}_\epsilon(M)$ indexed by the real numbers $\epsilon \geq 0$ defined as follows:

A subset $\{x_0, x_1, \ldots, x_k\} \subset M$ forms a k-dimensional simplex in $\mathrm{VR}_\epsilon(M)$ if and only if we have $d(x_i, x_j) \leq \epsilon \text{ for all } i, j$

Definition. Let M be a finite subset of a metric space (Z, d). The Cech filtration of M with respect to Z is the “increasing family" of simplicial complexes $\mathrm{C}_\epsilon(M)$ indexed by the real numbers $\epsilon \geq 0$

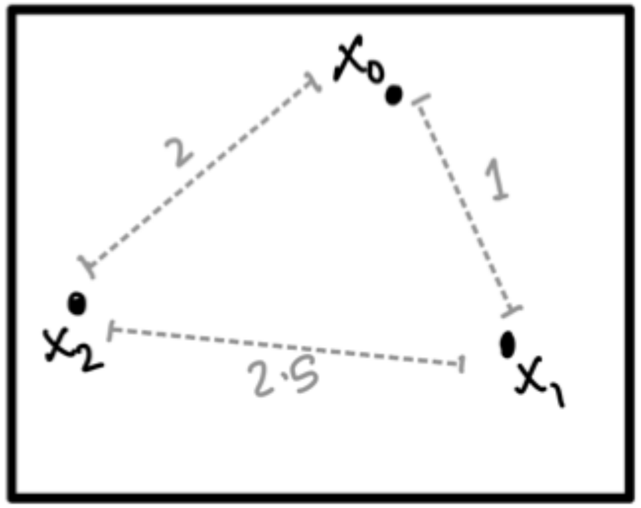
A subset $\{x_0, x_1, \ldots, x_k\} \subset M$ forms a k-dimensional simplex in $\mathrm{C}_\epsilon(M)$ if and only if there is $z \in Z$ with $d(z, x_i) \leq \epsilon \text{ for all } i$

Recap: Filtered simplicial complexes

Main example. Finite subset of points $M = \{x_0, x_1, \ldots, x_n\} \subset \mathbb{R}^d$

$\mathrm{VR}_\epsilon(M)$ is given by $d(x_i, x_j) \leq \epsilon$ for all i, j

$$\mathbf{VR}_\epsilon(M) = \begin{cases} \{x_0, x_1, x_2\} & 0 \leq \epsilon < 1 \\ \{x_0, x_1, x_2, x_0x_1\} & 1 \leq \epsilon < 2 \\ \{x_0, x_1, x_2, x_0x_1, x_0x_2\} & 2 \leq \epsilon < 2.5 \\ \{x_0, x_1, x_2, x_0x_1, x_0x_2, x_1x_2, x_0x_1x_2\} & \epsilon \geq 2.5 \end{cases}$$



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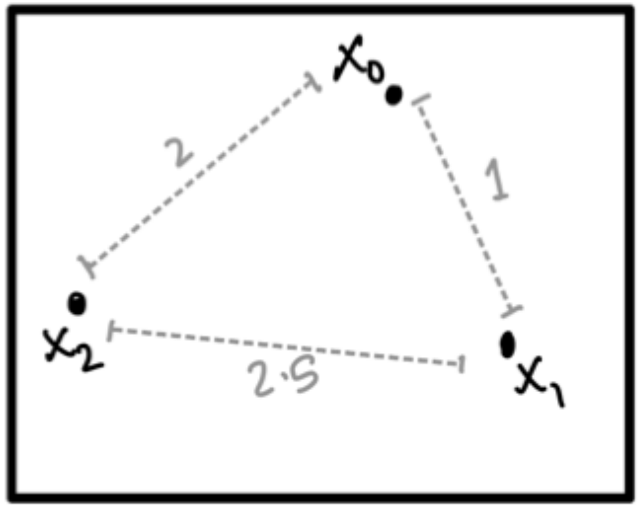
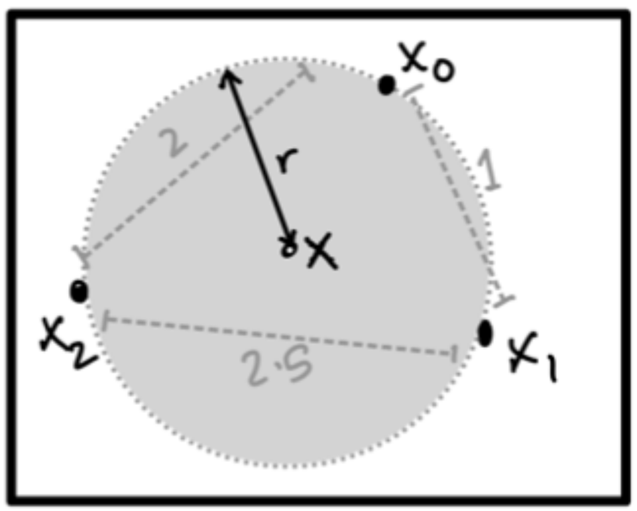


$\mathrm{C}_\epsilon(M)$ is given by $z \in Z$ with $d(z, x_i) \leq \epsilon$ for all i

$$\mathbf{C}_\epsilon(M) = \begin{cases} \{x_0, x_1, x_2\} & 0 \leq \epsilon < 0.5 \\ \{x_0, x_1, x_2, x_0x_1\} & 0.5 \leq \epsilon < 1 \\ \{x_0, x_1, x_2, x_0x_1, x_0x_2\} & 1 \leq \epsilon < 1.25 \\ \{x_0, x_1, x_2, x_0x_1, x_0x_2, x_1x_2\} & 1.25 \leq \epsilon < r \\ \{x_0, x_1, x_2, x_0x_1, x_0x_2, x_1x_2.x_0x_1x_2\} & \epsilon \geq r \end{cases}$$



Computational Algebraic Topology by Vidit Nanda

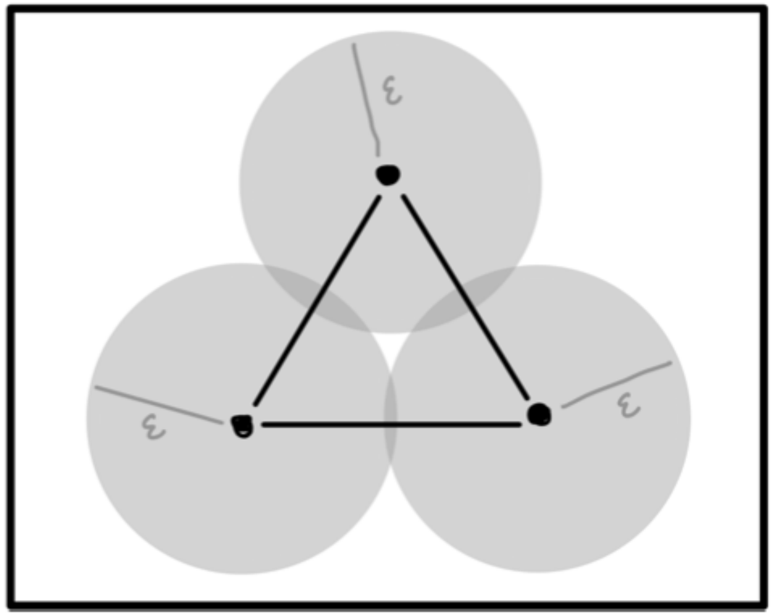
Recap: Filtered simplicial complexes

Algebra Geometry & Quantum Fields

Conclusions. Finite subset of points $M = \{x_0, x_1, \ldots, x_n\} \subset \mathbb{R}^d$

1. The Vietoris-Rips complex is a lot easier to compute — we only require the pairwise distances, which are given

2. However, the Cech complex is more faithful to the geometry of the union of balls which we sought to approximate, e.g.:



Exercise. Show $\mathrm{C}_\epsilon(M) \subset \mathrm{VR}_{2\epsilon}(M) \subset \mathrm{C}_{2\epsilon}(M)$, regardless of Z

Statistical vectorisation

Definition. Given the barcode $B = \mathrm{Bar}(V_\bullet, a_\bullet)$ of a tame PM with multiplicity $\mu\colon B \to \mathbb{Z}_{>0}$ and an interval [p,q], we call p the *birth*, q the *death*, and the length q-p the *lifespan* of the bar, respectively.

Definition. The persistence statistics vector of $\mu\colon B \to \mathbb{Z}_{>0}$ is

- The mean, standard deviation, median, interquartile range, full range, 10th, 25th, 75th, and 90th percentiles of the births p, the deaths q, the midpoints (p+q)/2, and lifespans q-p for all intervals [p,q] counted with multiplicity,
- Total number of bars (with multiplicity),
- The entropy $E_\mu = -\sum_{[p,q]\in B} \mu_{p,q} \left(\frac{q-p}{L_\mu}\right) \log\left(\frac{q-p}{L_\mu}\right) \in \mathbb{R}$

$$L_\mu = \sum_{[p,q]\in B} \mu_{p,q} \cdot (q-p)$$

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Statistical vectorisation

Definition. Given the barcode $B = \mathrm{Bar}(V_\bullet, a_\bullet)$ of a tame PM with multiplicity $\mu\colon B \to \mathbb{Z}_{>0}$ and an interval [p,q], we call p the *birth*, q the *death*, and the length q-p the *lifespan* of the bar, respectively.

Definition. The entropy summary function of $\mu\colon B \to \mathbb{Z}_{>0}$ is

$$S_\mu(t) = -\sum_{[p,q]\in B} \mathbb{1}_{p\le t\le q} \cdot \mu_{p,q} \left(\frac{q-p}{L_\mu}\right) \log\left(\frac{q-p}{L_\mu}\right) \in \mathbb{R}$$

where $L_\mu = \sum_{[p,q]\in B} \mu_{p,q} \cdot (q-p)$ as before. Note that $S_\mu\colon \mathbb{R}_{\ge 0} \to \mathbb{R}$

In practice, we choose, e.g., 100 equally spaced values of the parameter t, and record the function as an object in $\mathbb{R}^{100}$.

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Statistical vectorisation: Entropy

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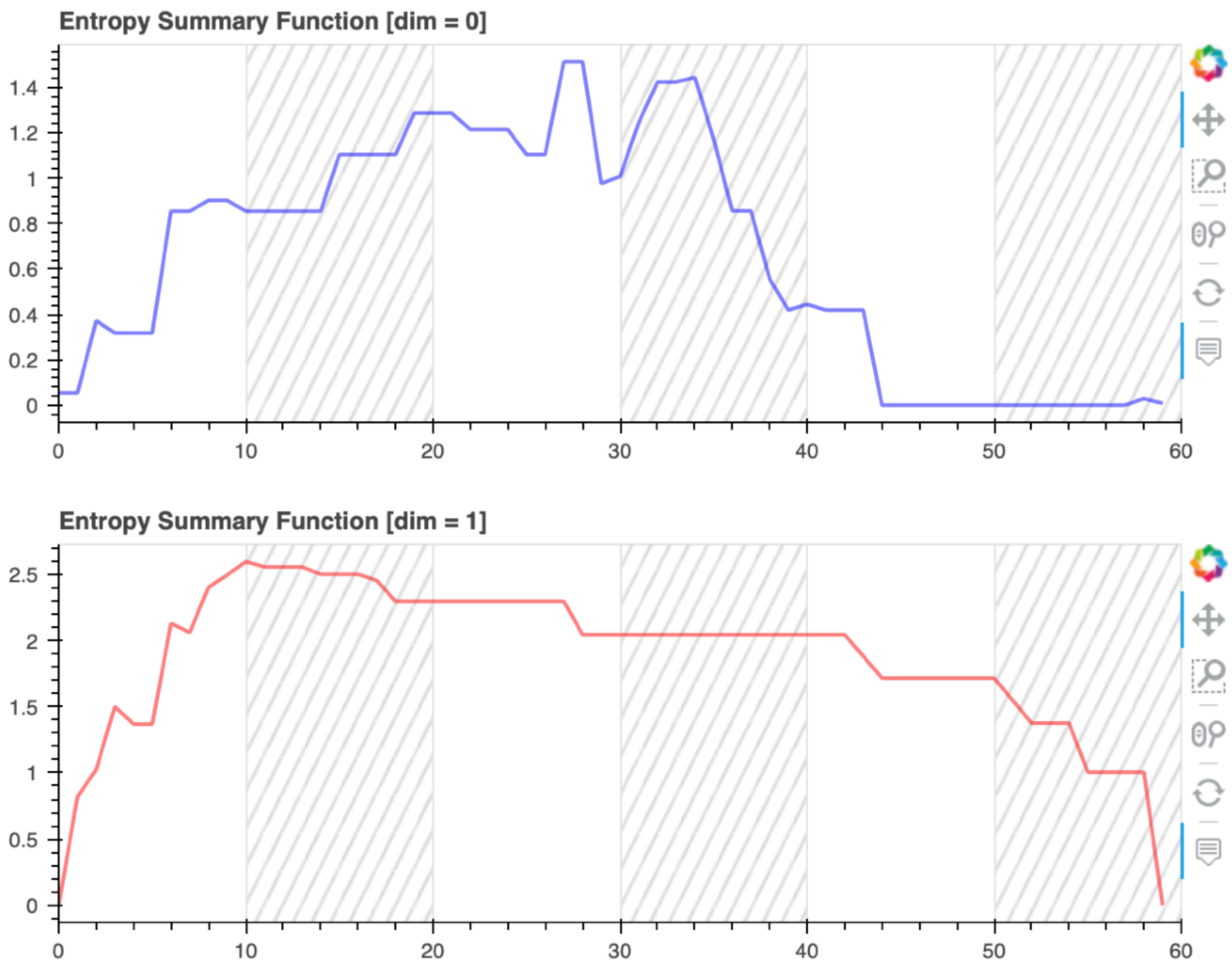

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Statistical vectorisation

In practice, we choose, e.g., 100 equally spaced values of the parameter t, and record the function as an object in $\mathbb{R}^{100}$.

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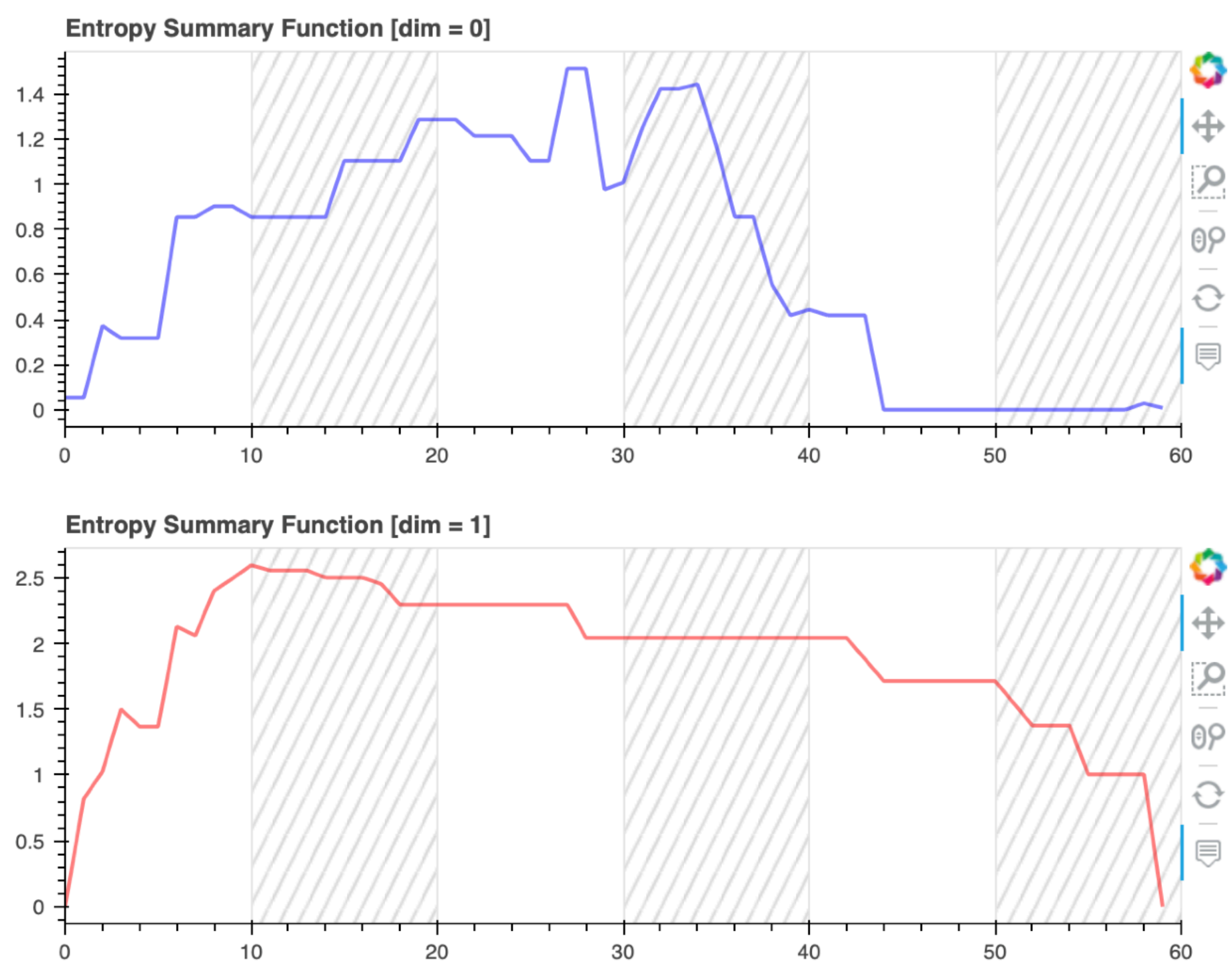

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